Exp no: 5
Date: 24/2/26

Verification of Norton's theorem.

Aim:

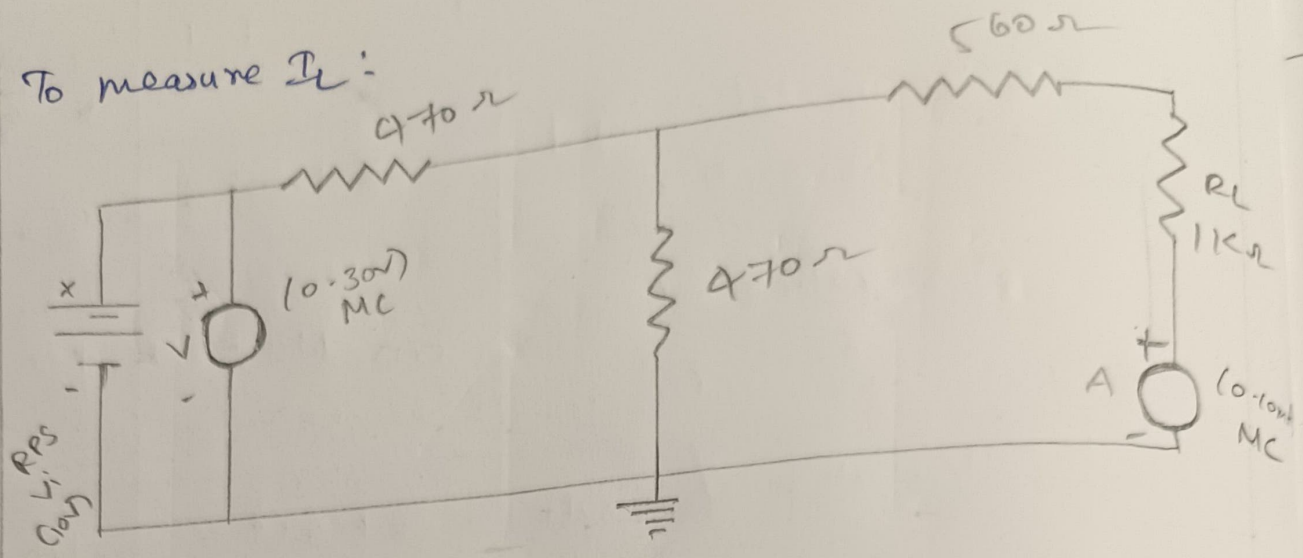
To verify Norton's theorem practically and theoretically for the given DC circuit.

Apparatus Required:

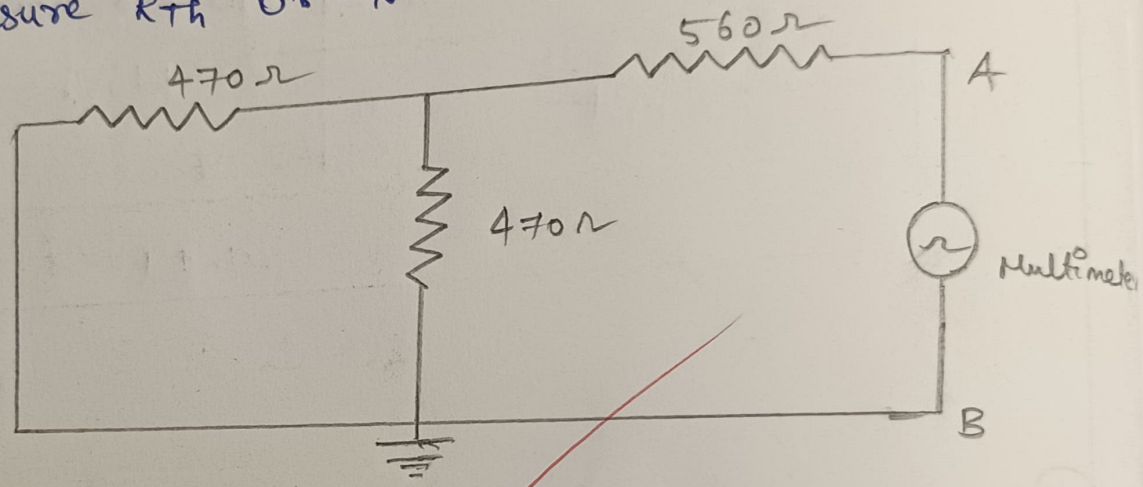
S.No	Apparatus	Specification	Qty.
1.	Regulated Power Supply (RPS)	(0-30)V	1
2.	Voltmeter	(0-30V) MC	1
3.	Ammeter	(0-10mA) MC	1
4.	Resistor	470Ω 560Ω 1KΩ	2 1 1
5.	Bread Board	-	1
6.	Multimeter	-	1

Circuit diagram :- Verification of Norton's theorem.

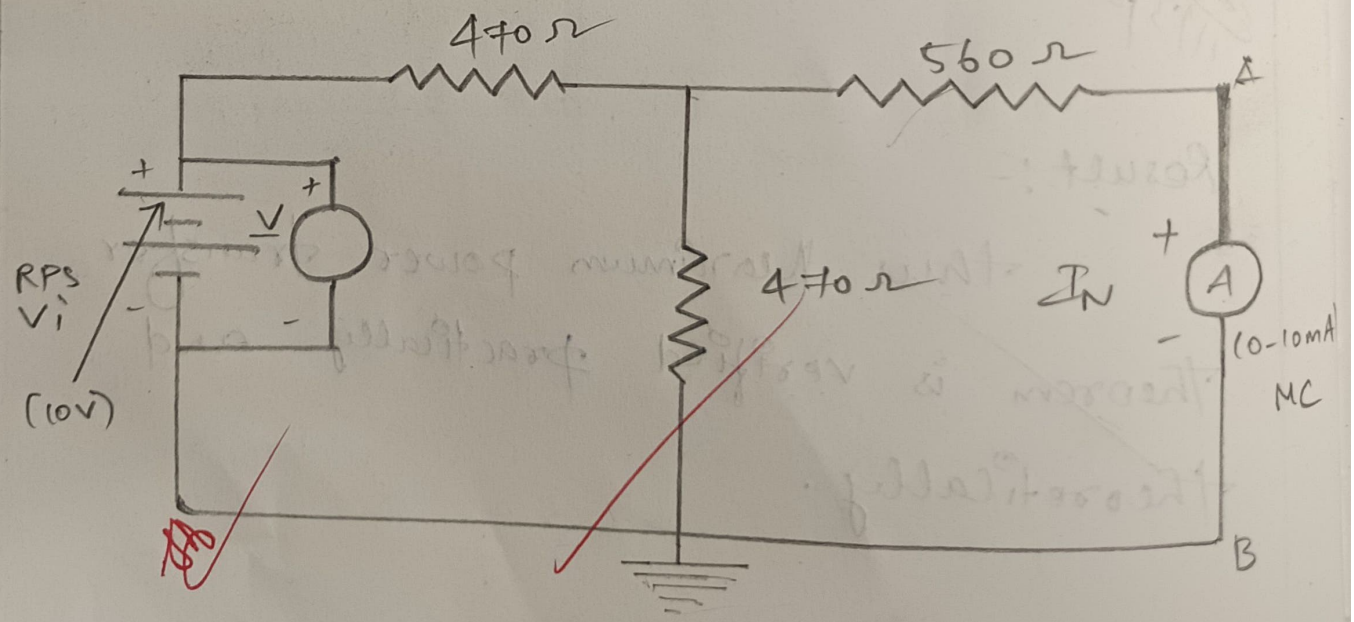
To measure I_L :



To measure R_{Th} or R_N



To measure I_N or I_{SC} :



theory :-

Norton's theorem :-

Norton's theorem states that, 'a linear two-terminal circuit can be replaced by an equivalent circuit consisting of current source, $I_N (= I_{sc})$ in parallel with a resistor $R_N (= R_{Th})$, where $I_N (= I_{sc})$ is the short-circuit current through the load terminals and R_N is the equivalent resistance at the load terminal when the independent source are turned off.

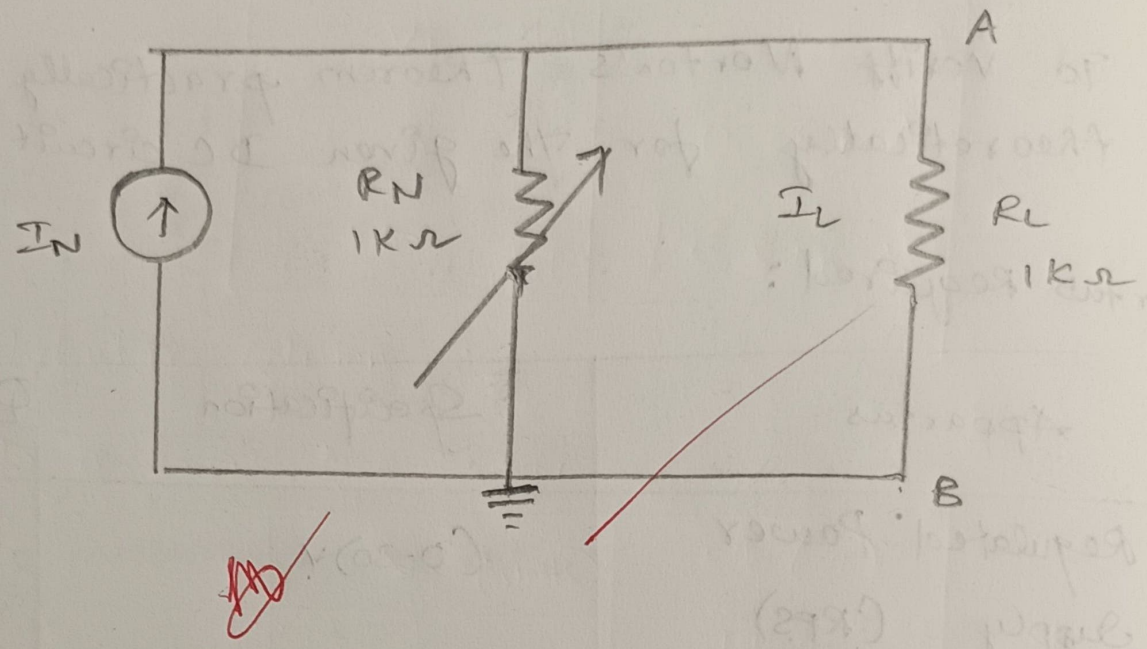
Norton's Current, I_N or I_{sc}:

It is the short-circuit current through the load terminal. i.e., $I_N = I_{sc}$

Norton's Resistance, R_N;

It is the look-back resistance across the load terminals when all the sources are replaced by their internal resistance. An ideal voltage source is replaced by short-circuiting as its

Norton's equevalent circuit;



internal resistance is zero. An ideal current source is replaced by open-circuiting as its internal resistance is infinity.

Tabular column : 1

To measure I_L

Vi (volts)	I_L (amps)
2	0.35 mA
10	5.51 mA

Tabular column : 2

To measure R_{TH} or R_N

Vi (volts)	R_{th} (Ω)
2	790 Ω
4	798 Ω
10	800 Ω

Tabular column : 3

To measure I_N or I_{SC}

Vi (volts)	I_N (amps)
10V	4.05
2V	0.80

O/P verified

Procedure:

1. Make the connection as per the circuit diagram.
2. Vary the RPS and set an input voltage of 10 V.
3. Note down the voltmeter reading (V_i) and Ammeter reading (I_L) in Tabular Column 1.
4. Switch off the supply and make connection for circuit diagram 2.
5. Measure the thevinin's theorem resistance R_{Th} = Norton's resistance R_N.
6. Switch off the supply and make connection for circuit diagram 3.
7. Set an input voltage of 10V in the RPS and Note down the voltmeter reading V_i and V_{th} (= V_{oc}) in tabular 3.
8. Switch off the supply and connection for circuit diagram 4.
9. Set an input voltage of 10V in the RPS and note down the voltmeter reading V_i and Ammeter reading I_N (= I_{sc}) in Tabular Column 4.

Model Calculation:

Practical value of $I_L = 2.3\,mA$

Verification of Norton's theorem:

$$I_L = I_N * R_N / (R_N + R_L) = 2.43\,mA$$

theoretical calculation of I_L, I_N and R_{Th} (R_N) for the given circuit:

$$R_{Th} = 470 + \frac{470 \times 560}{470 + 560} = 470 + 255.5$$

$$R_{Th} = 725.53\,\Omega$$

$$I_T = 10/725.35 = 0.013\,A$$

By CDR,

$$I_N = 0.013 \times \frac{470}{470 + 560}$$

$$= 0.013 \times 0.45 = 0.0059\,A$$

$$= 5.9\,mA$$

$$R_N = 560 + \frac{470 \times 470}{470 + 470} = 560 + 2.35 = 795$$

$$V_L = 2.61 \times 10^{-3}$$

$$V_L = 2.61\,V$$

$$I_L = I_N \times \frac{R_N}{R_N + R_L}$$

$$= 5.9 \times 10^{-3} \times 795 / 795 + 1000$$

$$= 2.01\,mA$$

10. Draw the thevinin's equivalent circuit and Norton's equivalent circuit as shown in circuit diagram 5 & 6 respectively.

11. Calculate the I_L value using the formula

thevinin's theorem

$$I_L = V_{Th} / (R_{Th} + R_L)$$

Norton's theorem

$$I_L = I_N * R_N (R_N + R_L)$$

12. theoretically verify the Norton's theorem.

Mark. Split-up -

S.No	Description	Marks Allowed	Mark obtained
1.	Experimental setup / Connection	20 Marks	20
2.	Observation / Reading	20 Marks	20
3.	Calculation / Result	20 Marks	20
4.	Viva	20 Marks	20
5.	Record.	20 Marks	20
		Total	100

10

11/8/26

Result:-

thus Thevinen's and Norton's theorem is verified practically and theoratically.